

JINDAL STAINLESS

ENGINEERING CASE STUDY COMPETITION 2026

PROBLEM STATEMENTS & GUIDELINES

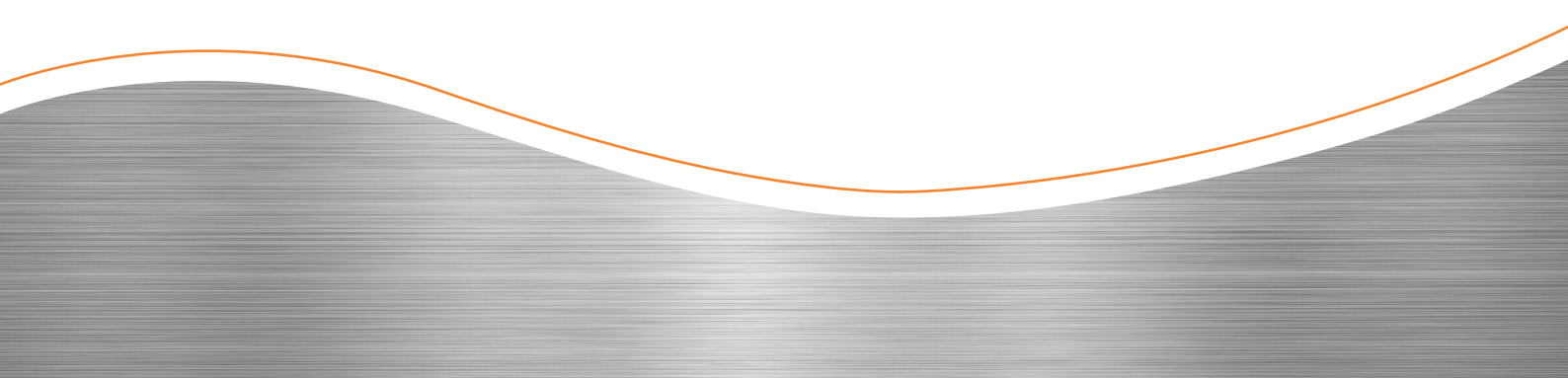
Innovation • Sustainability • Digital Manufacturing • Engineering Excellence



About Jindal Stainless

Jindal Stainless is India's largest stainless steel manufacturer and one of the world's leading stainless steel producers, with an annual melt capacity of over 3 million tonnes. The Company manufactures a comprehensive portfolio of stainless steel flat and long products catering to infrastructure, railways, automotive, architecture, process industries, defence, renewable energy, consumer goods and precision engineering.

With integrated manufacturing facilities at Jajpur and Hisar, Jindal Stainless combines advanced manufacturing, innovation and sustainability to deliver world-class products across global markets while creating value for customers through quality, technology and responsible manufacturing.



Problem Statement 3

Carbon and Energy Calculator for Steelmaking

Business Context

Sustainability and responsible manufacturing are integral to Jindal Stainless' long-term growth strategy. Increasing transparency around carbon emissions and energy consumption can help accelerate informed decision-making and support the industry's transition towards lower-carbon steelmaking.

Background

Stainless steel making is energy- and emissions-intensive, and carbon intensity per tonne depends heavily on the input mix and energy sources used. With a Net Zero by 2050 commitment, there is a need for simple tools that make the emissions impact of different choices visible and help identify lower-carbon options.

Action Item

Build a tool that estimates carbon emissions per tonne of steel based on a chosen input mix (for example share of scrap versus virgin input) and energy source and let the user adjust these inputs to see the effect on emissions. The tool should help users find combinations that minimise CO₂ per tonne while remaining practical.

Key Considerations

- Reasonableness of the emissions model and assumptions
- Clarity of how inputs affect the result
- An interactive, easy-to-understand interface
- The usefulness of the optimisation or comparison feature
- A measurable reduction in estimated emissions

Case Guidelines

Each participating team should submit a presentation comprising 4–5 slides that clearly communicates the proposed solution. The deck should be structured to demonstrate technical depth, business relevance and implementation feasibility.

Problem Understanding: Define the problem, explain its significance and establish the need for the solution.

Objectives: State the expected outcomes and measurable success indicators.

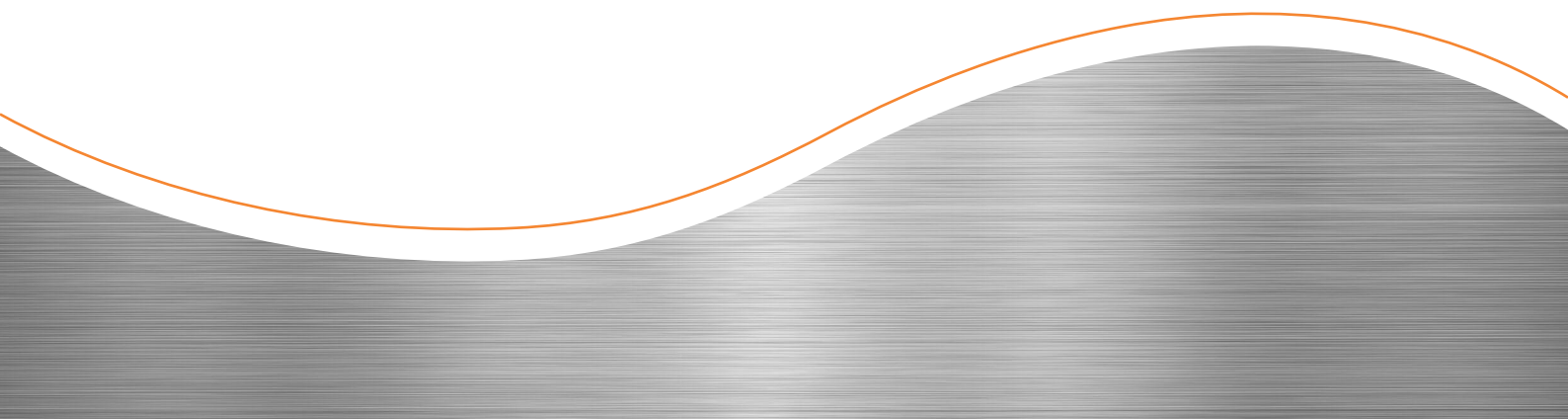
Proposed Solution: Describe the solution, research done, methodology, architecture, workflow, implementation plan and expected outcomes.

Validation & Feasibility: Present supporting evidence through calculations, simulations, datasets, prototypes or proof of concept.

Assumptions & Limitations: Highlight assumptions and implementation constraints.

Innovation & Differentiation: Explain how the solution improves upon existing approaches.

Expected Impact: Summarise the technical, operational, commercial and sustainability benefits.



Evaluation Criteria

Evaluation Parameter	Assessment Focus
Problem Understanding	Depth of understanding of the challenge.
Innovation & Originality	Novelty and creativity of the idea.
Technical Excellence	Engineering robustness and feasibility.
Business Relevance & Impact	Potential value to Jindal Stainless and customers.
Feasibility & Scalability	Industrial applicability and scalability.
Presentation & Communication	Clarity, structure and effectiveness.

